

- $X_d(i)$ - d-th coordinate of i-th particle's position
- $V_d(i)$ - d-th coordinate of i-th particle's velocity
- ω - Inertia weight
- $P_d(i)$ - d-th coordinate of i-th particle's *personal* best
- $G_d(i)$ - d-th coordinate of the globally (sometimes also only locally) best solution found
- $c1$ and $c2$ - Two weight values to balance exploiting the particle's best $P_d(i)$ and swarm's best $G_d(i)$
- $r1$ and $r2$ - Two random values being create for the velocity update

The velocity update is given by:

$$V(i)_{d+1} = \omega V_d(i) + c1r1(P_d(i) - X_d(i)) + c2r2(G_d(i) - X_d(i))$$

The corresponding position value is then updated by:

$$X(i)_{d+1} = X(i)_d + V(i)_{d+1}$$